

Batch Gradient Ascent for Logistic Regression — Exam Short Answers

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1. Algorithm — What & Why

Goal: MLE of logistic regression $\hat{\beta}$ via **gradient ascent** (not `glm()`).

- Model: $\pi_i = P(y_i = 1|x_i) = \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}}$
- Log-likelihood: $\ell(\beta) = \sum y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})$
- **Gradient (score):** $\nabla \ell(\beta) = X^\top (y - \pi)$ — concave in $\beta \Rightarrow$ unique max.
- **Update rule:** $\beta^{(k+1)} = \beta^{(k)} + t \cdot \nabla \ell(\beta^{(k)})$, t = step size.
- **Stop when:** $\|\nabla \ell(\beta)\| < \text{tol}$ (near-zero gradient = stationary pt = MLE, since concave).

2. Code — Line-by-Line (Short)

```
library(mcmc); data(logit)
y <- logit$y; X <- as.matrix(logit[, 2:5]); p <- dim(X)[2]
```

- logit data: 100 obs, response y, predictors x1–x4. **No intercept in X.**

```
f.gradient <- function(y, X, beta){
  beta <- matrix(beta, ncol=1)
  pi <- exp(X%*%beta)/(1+exp(X%*%beta))
  colSums(X*as.numeric(y-pi)) # = t(X)%*%(y-pi)
}
```

- Computes fitted $\hat{\pi}_i$, then $X^\top (y - \pi)$ via vectorized `colSums` (fast, no explicit transpose).

```
beta_k <- rep(0,p); t <- .1; tol <- 1e-8; iter <- 0; grads <- 100
while((grads>tol) && iter<1e4){
  iter <- iter+1
  foo <- f.gradient(y,X,beta_k)
  grads <- sqrt(sum(foo^2)) # ||gradient||
```

```

beta_k <- beta_k + t*foo          # ascent step
}

```

- **Init** $\beta^{(0)} = 0$. Each pass: get gradient $\rightarrow$ norm $\rightarrow$ step $\rightarrow$ repeat.
- **Why grads>tol**: loop runs *until* gradient ≈ 0 (found the max).
- **Why iter<1e4**: safety cap if it never converges.
- store.beta, store.grads just log history for the plot — no algorithmic role.

3. Output — Values & Meaning

```

## [1] 27
##          x1          x2          x3          x4
## 0.7178916 0.8789924 0.4227914 0.5874717

```

- **Iterations to converge**: 27
- **MLE**: $\hat{\beta} \approx (0.718, 0.879, 0.423, 0.587)$ for x_1, x_2, x_3, x_4 — matches `glm(y~.-1,binomial)` exactly (independent check).
- **Interpretation**: e.g. $\hat{\beta}_2 \approx 0.879 \Rightarrow$ 1-unit $\uparrow$ in x_2 multiplies odds of $y = 1$ by $e^{0.879} \approx 2.41$ (others fixed).

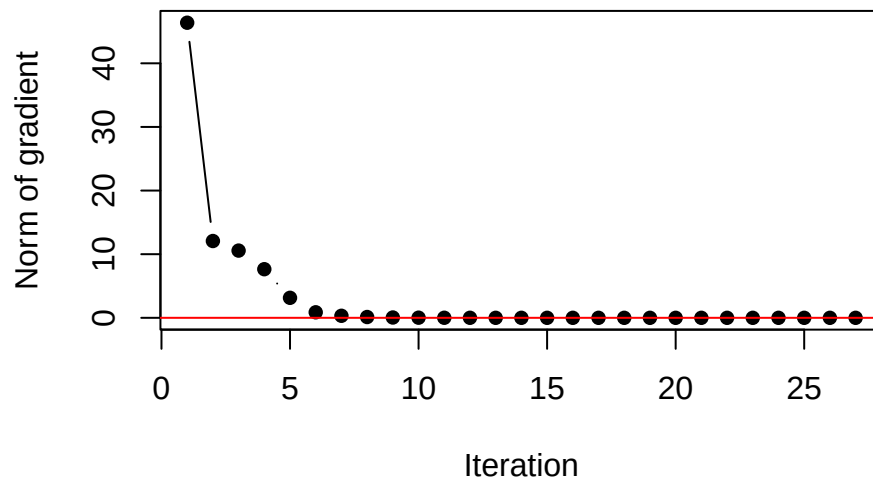


Figure 1: Gradient norm $\rightarrow 0$: fast drop, then flat (concave objective, converged).

- **Plot shape**: steep fall (iters 1–6) $\rightarrow$ flattens near 0 by $\sim$ iter 10 $\rightarrow$ satisfies tol at 27. Classic concave-ascent convergence.

4. One-Line Recap (memorize)

“Gradient ascent maximizes logistic log-likelihood by repeatedly stepping $\beta \leftarrow \beta + t X^\top (y - \pi)$ until the score is ~ 0 ; here it converges in 27 steps to $\hat{\beta} \approx (0.72, 0.88, 0.42, 0.59)$, matching `glm()`.”